**UGC Major Research Project Summary Report**

**ENTITLED**

# DEVELOPMENT OF NOVEL GRAPHENE AND METAL NANO COMPOSITE FILMS AND CHARACTERIZATION FOR LABEL FREE ELECTROCHEMICAL DNA-PROTEIN SENSING

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**TO**

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Graphene, a two-dimensional sheet structure of $sp^2$ conjugated atomic carbon, has attracted considerable attention in a myriad of research works due to its outstanding electronic, electrochemical, and thermal properties. Applications of graphene transducers include constructions of transistors solar-cells, supercapacitors, and sensors. Excellent biocompatibility facilitates interaction with DNAs, aptamers, peptides, enzymes and proteins. The DNA and proteins have been studied on individual graphene transducer and immobilized directly by nonpolar or aromatic ring π-electron interactions and not studied the complexation of DNA with protein. Further, the non-covalent π-interaction results in poor performance of the sensor and data instability. Recently, we have reported covalent immobilization of DNA on graphene –AuNP composite for sensitive detection, however, the graphene is functionalized with dendrimer and studied DNA hybridization. This step needs to be eliminated as the graphene oxide possesses large number of functional groups amenable for bio functionalization. One important goal of practical application is to develop sensing platform with high discrimination effect. Since metal nano particles are well reduced on GO, enhances conductivity, detection sensitivity, in this project, preparation of DNA –protein complex on the graphene –metal nano particle platform is proposed.

**Objective of the Project**

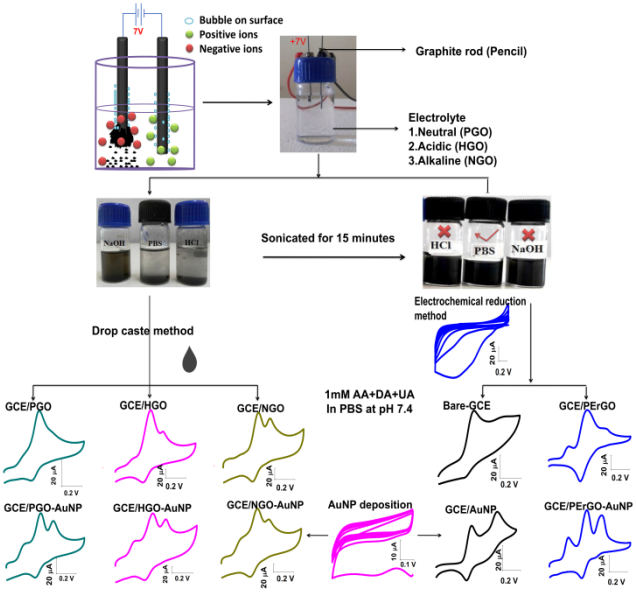
- **First objective** is to develop stable graphene-metal nano composites, characterization and optimization.
- **Second objective** is construction of an efficient DNA monolayer with minimized non-specific interaction of DNA and to maximize the distance between the each capture probes for achieving maximum hybridization efficiency and discrimination efficiency between the single and double stranded DNAs formed on the graphene-metal nano particle of different composites.

- **Third objective** is to study the interaction of cancer DNA for efficient p53 antibody binding. Control experiment will be made in presence P21 antibody sequence that would not interact with damaged cancer DNA sequence.
- **Fourth objective** is to study the charge transport through complementary target hybridized surfaces in presence of the electroactive probes and double stranded specific cationic intercalators to understand the DNA mediated charge transport for DNA-protein sensors

## WHETHER OBJECTIVES WERE ACHIEVED

### Plan 1

**Preparation and characterization of different graphene-metal nano particle composites and characterization and optimization**




**Scheme1:** Methodology proposed for fabricating different reduced graphene oxide for sensor development.

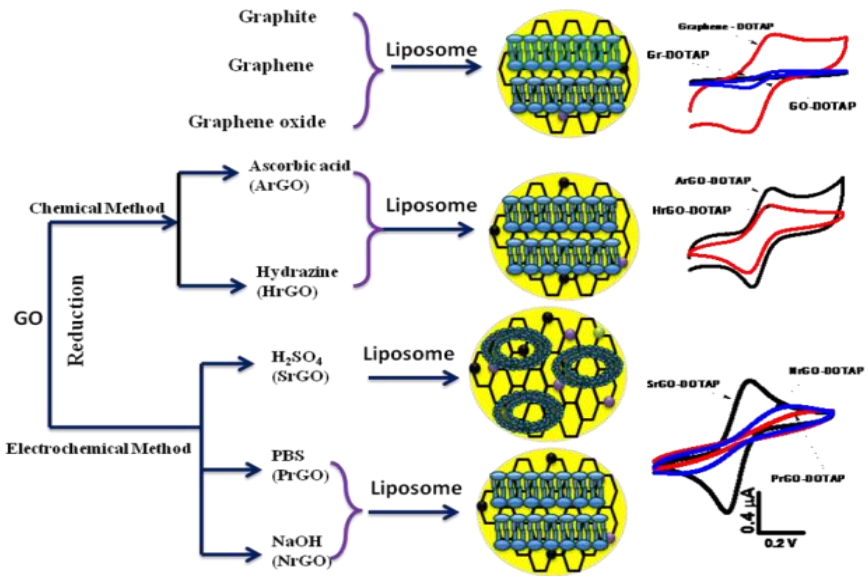
Pencil graphite is electrochemically exfoliated to obtain few layered graphene oxide (GO) in hydrochloric acid (HCl, HGO), sodium hydroxide (NaOH, NGO) and phosphate buffer saline (PBS, pH 7.4, PGO) media at a constant potential (+7.0 V) without ionic liquid. Thus, the obtained graphene oxide is deposited (by drop casting and electro reduction) onto a

glassy carbon electrode (GCE). Following this, gold nanoparticles (AuNP) are electro deposited to form GO–AuNP composite and characterizations were made. The surface was applied for electrochemical sensing of neuro transmitters as the model systems. **The results have been published in the RSC Advances 2015.**

## Plan 2

**Construction of an efficient DNA monolayer with minimized non-specific interaction of DNA and to maximize the distance between the each capture probes for achieving maximum hybridization efficiency and discrimination efficiency between the single and double stranded DNAs formed on the graphene-metal nano particle composite**.

Graphene based sensors have been reported for direct DNA sensing and DNA are claimed to interact with graphene by hydrophobic interaction. This limits the sensitivity and immobilized DNA could not be used for further applications as they lose their electronic properties. In this context, covalent attachment of DNA has been reported and to minimize the experimental procedure in a cost effective way, graphene metal nanoparticle systems have been reported. However, only few nanoparticles are attached on the graphene. Recently, graphene oxide attracts greater interest for anchoring metals. Still the sensitivity of the system is insufficient to detect trace quantity of DNA and protein targets.



**Scheme 2:** Proposed scheme of Interaction between DOTAP liposome on different graphene materials synthesized by chemically using ascorbic acid (ArGO) and hydrazine (HrGO) reductants and electrochemically in sulphuric acid (srGO), phosphate buffer (prGO) and sodium hydroxide (NrGO).

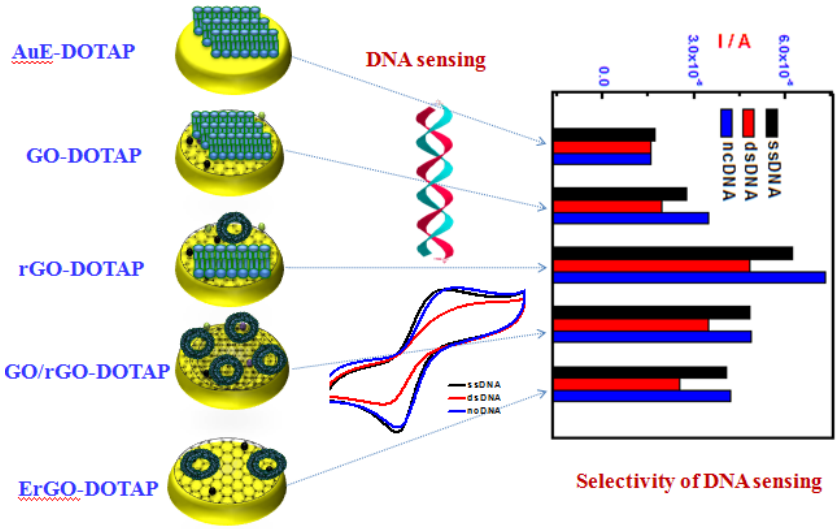
To improve the sensitivity further, liposome, which form bilayer on solid surfaces, may provide large active centers for metal nano particle anchoring are used. In this, intact liposome vesicle is tethered on the reduced graphene oxide without the assistance of gold nanoparticle. There are no literatures available reporting the formation of spherical liposome without the help of gold nanoparticle. Direct immobilization of DNA on the rGO-DOTAP and target DNA detection results in poor sensitivity and selectivity due to uncontrolled orientation of the single stranded DNA. For the covalent immobilization of DNA on the rGO-DOTAP surface, it is modified with AuNPs through the electrostatic interaction between the DOTAP head group and citrate capped AuNP. Now, the thiolated DNA is immobilized through the gold–thiol chemistry on the rGO-DOTAP–AuNP and applied for electrochemical label free DNA detection. This was resulted in higher discrimination effect between ssDNA and dsDNA in presence of electro active $[Fe(CN)6]^{3-/4-}$. **The results are published in Sensors and Actuators B: Chemical 2018.**

## Plan 3

**Development of spherical lipid-nanoparticle on graphene oxide/reduced graphene oxide composite for label free DNA sensing**

Controlling the vesicle to bilayer structure of cationic DOTAP on the GO is defiant due to the existence of large electrostatic forces which drives the vesicle fusion and bilayer formation on the GO. On the other hand, high hydrophobic of the graphene also drives the vesicle fusion and bilayer formation. Hence the amount of surface functional groups and hydrophobic character of the GO has to be controlled. In this approach, the traditional physical mixing of chemically reduced GO with the unreduced GO is attempted to form a

trilayered GO/rGO network with the lowered oxy functional groups and hydrophobic. For this the GO was prepared by the conventional modified Hummers method and reduced using mild reductant ascorbic acid. The interactions of liposome vesicles are examined by investigating changes in the electrochemical signals of the rGO/GO/rGO thin film electrodes in presence of electro active redox probes. For demonstrating the application of these GO supported liposome vesicles in drug loading and molecular sensing, the GO supported liposome is further protected using colloidal gold nanoparticles and utilized for label free DNA sensing. The results are submitted to the journal of electroanalytical chemistry, 2018.



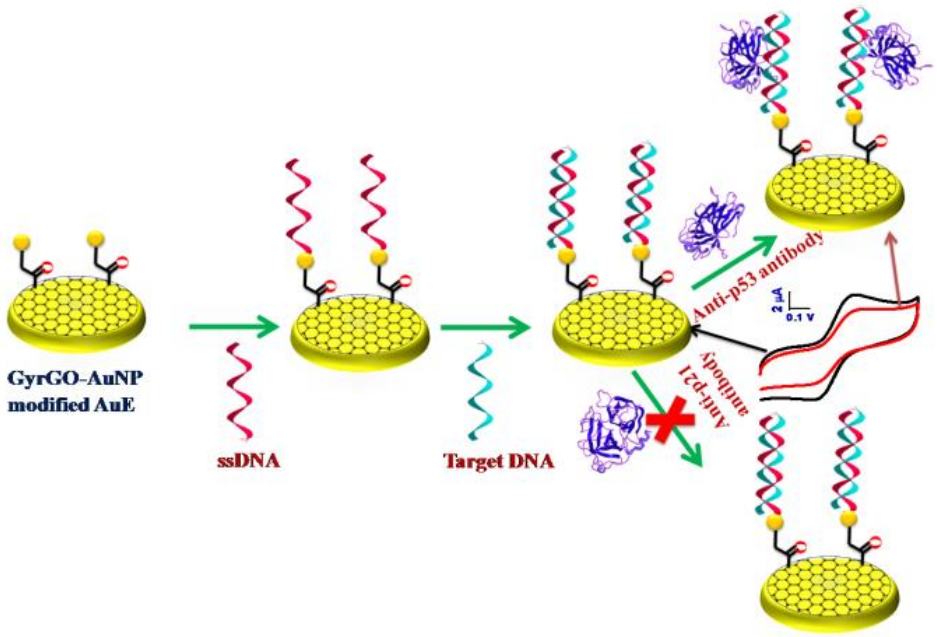

**Scheme 3 S**elective detection of label free electrochemical DNA sensing and lowest detection limit observed for DOTAP vesicle modified graphene surfaces.

## Plan 4

**To study the interaction of cancer DNA for efficient p53 antibody binding. Control experiment will be made in presence P21 antibody sequence that would not interact with damaged cancer DNA sequence**

**To study the charge transport through complementary target hybridized surfaces in presence of the electroactive probes and double stranded specific cationic intercalators to understand the DNA mediated charge transport for DNA-protein sensors**

Construction and characterization highly reproducible dsDNA sensor decorated graphene – gold nanoparticle. This plat form is used to understand the interaction of P53 antibody (protein).




**Scheme 4** Selective label free DNA-anti-p53 protein sensing at g-G-AuNP sensor surface by electrochemical method.

Detection of p53 antibody, an universal cancer biomarker, essential for the early diagnosis and prevention of cancer mortality. This is limited due to the poor sensitivity of the existing electrochemical sensor platforms. Here, we prepared graphene by electrical exfoliation of pencil graphite in presence of glycine in aqueous solution without using any intercalting agents. Graphene has been decorated with gold nanoparticles prior to the biofunctionalization with double strnaded DNA sequence, representing lung cancer, followed by its hybridization with a universal cancer biomarker anti-p53 antibody (cancer biomarker). Change in the volatmetric signal of the graphene before and after these molecular binding events is monitored in the presence of ferri/ferro cyaninde redox probe. The binding os DNA antibody has been theoretically confirmed using bioinformatic tool. Selectivity of the sensor is demonstrated using anti p21 antibody and DNA sequences from *E-Coli* pathogen. The

cancer biomarker antibody can be detected in the concentration range from 0.1 ng/L to 0.1 µg/L. The minimal concentration of dsDNA required for the efficient sensing of anti-p53 antibody is 0.6 fM. Both graphene preparation and cancer biomarker sensing methods are simple, rapid, easy to fabricate, stable and can be utilized for practical applications. **The results are submitted to Analytica Chemica Acta, 2018.**

## SUMMARY OF THE FINDINGS

Graphene oxide and reduced graphene oxides were prepared by chemical (Hummer's method) and direct electrical exfoliation methods. Mild chemical reductant ascorbic acid and electrochemical potential cycling of graphene oxide were used further for preparing reduced graphene oxide. The samples were characterized by Ultraviolet Visible (UV-Vis), Photoluminescence (PL), X-ray diffraction (XRD), Fourier transform Infrared (FTIR), contact angle goniometry (CA) and scanning electron microscopy (SEM) techniques. These materials are further anchored on the gold electrode by simple drop casting followed by their interaction with prepared cationic liposome (eg., DOTAP) is investigated by electrochemical method to identify the vesicle and bilayer structure. Comparative study of cationic DOTAP lipid interaction with these graphene surfaces indicate the fact that the thin layered rGO produced through mild electrochemical reduction in the acidic medium (favors selective reduction of hydroxyl and epoxide functional group) do not ruptures the intact DOTAP vesicle structure confirmed by different surface characterizing techniques AFM, TEM and QCM techniques. The results indicate that the presence of bilayer or intact vesicle structure is found to be highly depending on the extensiveness of reduction of functional groups on the GO surface. Graphene supported liposome structure is further decorated with gold nanoparticle which plays dual role of stabilizing the liposome vesicle structure and provides active centre for anchoring DNA probes by covalent linkage for DNA sensing. Application of this system for label free DNA detection by the electrochemical method gives the lowest

detection limit of $10^{-17}$ M with a linear range $10^{-15}$ to $10^{-6}$ M demonstrating the feasibility of the sensor for molecular diagnosis. Influence of GO preparation method has been studied by direct electrical exfoliation of pencil graphite in presence and absence of intercalative agent (in this study glycine) on their electroactivity and biosensing of DNA-antibody and dopamine.

**CONTRIBUTION TO THE SOCIETY**

Graphene-gold nanoparticle composite and the method of dopamine neurotransmitter sensing are useful for monitoring the dopamine level in the pharmaceutical industry. Similarly, method of controlling the interaction of liposome vesicle on the graphene surface could enable the development tools for monitoring not only the biomembrane - molecule (DNA, protein) sensing but also for efficient drug delivery applications